CHAPTER 4

Differentiation, convexity and continuity

S29, S31, S32; reviews in S33 and S40

Before you begin · Differentiation and reading graphs

Without a calculator, in six minutes. Answer briefly. Use the derivatives from Première and sign tables. Then choose the targeted follow-up indicated under the questions.

1. Differentiate x^3.

2. Differentiate $(x+1)e^x$.

3. Give the sign of $(x-2)(x+3)$ on $]-3;2[$.

4. What does $f'(a)$ graphically?

5. Give an equation of the tangent at a.

6. Give a function that is not differentiable at 0.

Answers

$3x^2$; $(x+2)e^x$; negative ; slope of the tangent ; $y = f(a) + f'(a)(x-a)$; $|x|$.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Use the rule for differentiating powers. **2.** Write the two terms of the derivative of the product separately; R35. **3.** Draw the sign table of both factors. **4.** Calculate a slope with two points; R37. **5.** Use the value and the slope at the point of contact; R37. **6.** Compare the slopes to the left and to the right of the absolute value. The R questions serve as review material with the teacher, even if their whole-class routine is planned later.

Objectives

Differentiate a composite; study variations and convexity; prove existence and uniqueness; approximate a solution.

Course notes

Differentiating a composite

If u is differentiable on I and v differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The domains must be checked before calculating.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad \left(\frac{1}{u}\right)' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ & $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines the variations of f; the sign of f does not determine them.

Convexity

A function f is convex on an interval I if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, we accept the equivalences: f convex; f' increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave function. An inflection point is a point where the curve changes convexity.

Proof: position relative to the tangents. Suppose $f'' \geq 0$. Fix a and let $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g reaches its minimum $g(a) = 0$ at a, therefore $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not the x-coordinate of an inflection point. x^3 changes convexity at 0. The convexity of the exponential function and its tangent at 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous at a if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

Intermediate value theorem

If f is continuous on $[a,b]$, any real between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is moreover strictly monotonic, this preimage is unique. We can also work on an open interval using the limits at the endpoints.

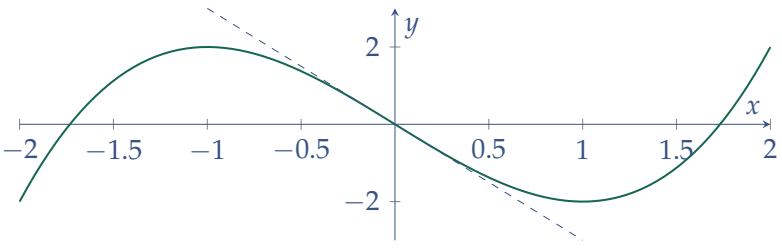
Worked example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Since $f(0) = -1$ and $f(1) = 1$, the IVT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0{,}68) < 0 < f(0{,}69)$ provides bounds, without replacing the justifications of existence and uniqueness.

A piecewise-defined function. For $x \neq 1$, set $g(x) = \frac{x^2-1}{x-1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The value at the point must therefore be compared with the limit, not replaced by it without justification.

Sequences and continuity. If $u_n \to \ell$ & f is continuous at ℓ, then $f(u_n) \to f(\ell)$. For a recursively defined sequence convergent $u_{n+1} = f(u_n)$, we get $\ell = f(\ell)$.

Bisection

For f continuous and $f(a)f(b) < 0$, calculate $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose the endpoints have images of opposite signs. After n steps, the length is $\frac{b-}{2^n}$. The midpoint gives an error of at most half the length. The method does not guarantee uniqueness; that is proved separately.



$y = x^3 - 3x$ and its tangent at the inflection point.

Linking calculation, curve and argument

4.1 The inner factor and the domain

First use: S29.

Worked example · A two-step composition

For $f(x) = \sqrt{1+x^2}$, the argument is always strictly positive: f is differentiable on $\mathbb{R}$. The outer function has derivative $t \mapsto 1/(2\sqrt{t})$, the inner function has derivative $2x$. Their product gives $f'(x) = x/\sqrt{1+x^2}$. The fraction is negative for $x < 0$, positive for $x > 0$: f decreases then increases.

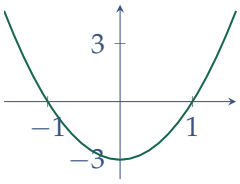
A1 · Immediate application Determine the domain of differentiability and the derivative of $g(x) = \sqrt{2x-1}$ and then $h(x) = e^{1-x^2}$. Check the sign of g'.

A2 · Fix a derivative "For $f(x) = (3x+1)^4$, we have $f'(x) = 4(3x+1)^3$." Explain the missing factor and suggest a check by expanding or using the rate of change at 0.

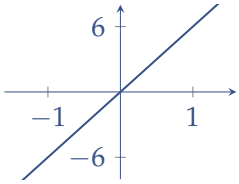
4.2 Match f, f' & f''

First use: S29 and S31.

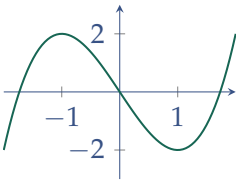
Curve A



Curve B



Curve C



Worked example · A positive decreasing function

For $g(x) = e^{-x}$, g is positive, $g' = -e^{-x} < 0$ and $g'' = e^{-x} > 0$. The curve goes down, but its negative slopes increase towards 0: it is convex. The three pieces of information are consistent.

A3 · Read before calculating The three curves correspond, in an order to be determined, to $f(x) = x^3 - 3x$, f' & f''. Match them using zeros, extrema and changes of convexity; then check by differentiating.

Property · Two readings not to be confused

$f'(a) > 0$ describes a positive slope of f, not a positive y-coordinate. $f''(a) > 0$ indicates locally that the slopes increase; this does not mean that f increases. For example e^{-x} decreases and remains convex.

4.3 Between a tangent and a chord

First use: S31; revisited in S33.

Worked example · Two bounds for x^2

On $[0;2]$, the chord joining $(0;0)$ to $(2;4)$ has equation $y = 2x$; the tangent at 1 has equation $y = 2x - 1$. Convexity gives 2 $x - 1 \leq x^2 \leq 2x$. The differences check out: $x^2 - (2x-1) = (x-1)^2 \geq 0$ and $2x - x^2 = x(2-x) \geq 0$ on this interval.

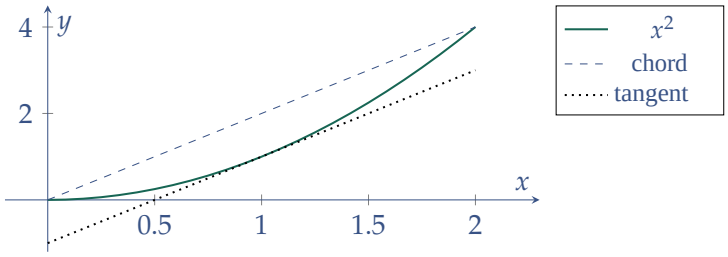


Figure 4.1. Compare the three y-values for $x = \frac{1}{2}$, then justify for every $[0;2]$.

A4 · Compare three functions For x^4, x^3 & $|x|$, study at 0 the continuity, differentiability and change of convexity. For both polynomials, explain why a vanishing second derivative is not enough to conclude.

4.4 Existence and uniqueness are two conclusions

First use: S32 and S33.

Worked example · Writing up an approximate solution

Let $F(x) = x^3 + x - 1$. It is a polynomial, so it is continuous. $F(0) < 0 < F(1)$ gives a solution in $]0;1[$ by the IVT. $F'(x) = 3x^2 + 1 > 0$ on $\mathbb{R}$ ensures that there cannot be two. The calculation $F(0{,}68) < 0 < F(0{,}69)$ locates this solution, but replaces neither existence nor uniqueness.

A5 · Two counterexamples Does the fn equal to -1 for $x < 0$ and 1 for $x \geq 0$ change sign without vanishing? Which hypothesis of the IVT is missing? For $x^2 = 1$, which hypothesis is missing to claim a unique solution on $\mathbb{R}$?

4.5 Keep information at every step

First use: S33 and S40.

Property · Bisection invariant

For a continuous function, maintaining $F(a)F(b) \leq 0$ preserves at least one zero in $[a;b]$. À At each step, the length is halved. The algorithm gives bounds; uniqueness must be established separately.

Worked example · A first interval

For $F(x) = x^2 - 2$ on [1;2], $F(1) = -1$, $F(2) = 2$. At the midpoint $m = 3/2$, $F(m) = 1/4 > 0$; the half that keeps the change of sign is $[1;3/2]$. Its width is 1/2.

A6 · Choice of strategy Prove that $e^{-x} = x$ has a unique solution in $[0;1]$. Perform two bisection steps, then determine how many steps guarantee an interval width of at most 1/128.

A first step: Reduce the equation to finding a zero of a function. The number of steps comes from the width, not of the values of the function.

One minute, spoken aloud. State the existence of a solution in one sentence containing the interval, the continuity and the values at the endpoints; state uniqueness in a separate sentence.

Delayed review · S47 then S66. Apply the same arguments to a logarithmic function, then to an area dependent on a bound.

Investigation

Derivatives and reading graphs - 35 min.

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Let $\mathcal{C}_f$ its graph.

1. Calculate $f'(x)$ and $f''(x)$, then draw their sign tables.
2. Deduce the variations of f and its intervals of convexity and concavity.
3. Determine the coordinates of the inflection point of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure the uniqueness of a solution.

Progressive exercises

E1 · Direct application. We define on $\mathbb{R}$ the functions $f(x) = (2x-1)^5$ & $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Application and justification. Let f the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals on which f is convex or concave.

E3 · Method to organise. Let $\mathcal{C}$ be the curve of the exponential function. Find an equation of its tangent at the point with x-coordinate 1. Using convexity, deduce an inequality valid for any real x.

E4 · Choice of strategy. Prove that the equation $x^3 + 2x = 5$ has a unique real solution. Bound this solution between two consecutive integers.

E5 · Synthesis or proof. A bisection algorithm starts from the interval $[0;2]$ and divides its length by two at each step. Determine the minimum number of steps guaranteeing an interval of length at most 10^{-3}.

Work from the textbook or a past paper

Barbazo, p. 132, Exercise 29. In class after studying convexity: questions 1 and 2. At home: finish questions 3 to 5. The aim is to distinguish the sign of f', which governs the variations, from that of f'', which governs convexity.

Homework

DM4 - A maximum and an equation - 60 min.

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its graph is denoted $\mathcal{C}_f$.

1. Calculate $f(0)$ and the limit of f at $+\infty$. Calculate $f'(x)$, then draw the table of variations of f.

2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of any inflection pt of $\mathcal{C}_f$.

3. Prove that the equation $f(x) = \frac{1}{2}$ has a unique solution α on $[0;+\infty[$.

4. Write a Python function that uses bisection on $[0;2]$ and returns two real numbers a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Is the position of the curve relative to this single tangent enough to assert that f is concave on its whole domain? Justify.

MCQ A · differentiation and convexity

After S29; the function is differentiable on each interval considered.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. If $f''(x) > 0$ on an interval, f is:

A. decreasing
B. convex
C. concave
D. constant

2. The derivative of e^{1+x^2} is:

A. e^{1+x^2}
B. $2xe^{x}$
C. $2xe^{1+x^2}$
D. $1+2x$

3. A tangent to the graph of a convex function is:

A. always above
B. always below on the interval
C. parallel to the axis
D. does not exist

4. If $f'(x)$ changes sign from $+$ to $-$ at a, then f :

A. has a minimum at a
B. has a local maximum at a
C. is convex
D. is not continuous

5. The function $x \mapsto |x|$ is:

A. differentiable at 0
B. not continuous at 0
C. continuous but not differentiable at 0
D. concave on $\mathbb{R}$

MCQ B · continuity and the IVT

After S33.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. To apply the IVT to $f(x) = 0$ on $[a; b]$, first check:

A. that f is continuous on $[a; b]$

B. that f' is zero

C. that $a = b$

D. that f is affine

7. If $f(a) < 0 < f(b)$ and f is continuous on $[a; b]$, then:

A. f has exactly one root

B. f has at least one root

C. f is increasing

D. $f(a) = f(b)$

8. To prove that a solution is unique, it is also enough, for example, that f be:

A. bounded

B. strictly monotonic

C. even

D. differentiable at a only

9. The bisection method preserves at each step:

A. two bounds of the same sign

B. an interval containing a root with opposite signs at the endpoints

C. a horizontal tangent

D. an exact value of the root

10. A graph of f'' below the axis means that f' is:

A. increasing

B. decreasing

C. positive

D. zero

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every continuous function has a root.

2. If f is convex, then f' is increasing where it is defined.

3. An approximate value obtained by bisection is a proof of existence.

Review of prior work

I can link the sign of f'', convexity and variations of f'; isolate, then justify a solution. **To watch out for:** claiming uniqueness without monotonicity; confusing a graphical conjecture with the IVT.

MCQ solutions

1. **B.** The sign $f'' > 0$ means that f' increases, hence the convexity. It does not give the sign of f' nor that of f.

2. **C.** The derivative of $e^{u(x)}$ is $u'(x)e^{u(x)}$. Here $u(x) = 1 + x^2$, therefore $u'(x) = 2x$. Forgetting this factor is a composition error.

3. **B.** A differentiable convex function lies above each of its tangents on the interval of convexity. These tangents need not be horizontal.

4. **B.** The positive sign of f' then negative gives an increase then a decrease around a: a local maximum, not a minimum.

5. **C.** $|x| \to 0$ when $x \to 0$, hence continuity. The left and right slopes are -1 and 1, so it is not differentiable at 0.

6. **A.** The IVT uses continuity on the whole interval. No zero derivative is required; a reduced interval does not give a change of sign.

7. **B.** Continuity and a change of sign guarantee existence. A function may have several roots: uniqueness requires an additional argument.
8. **B.** A strictly monotonic function is injective: a given value has at most one preimage. Being bounded or even does not guarantee this property.
9. **B.** As long as no exact root is found, we keep the sub-interval whose endpoints have images of opposite signs; its length is halved.
10. **B.** Over an interval where $f'' < 0$, the derivative of f' is negative, so f' decreases. Its sign remains to be studied.

True/false. False: e^x is never zero. True if f is differentiable on the interval: this is a characterisation of convexity. False: an approximate value alone does not prove existence; the bisection must start from an interval on which continuity and a change of sign are established.

Solutions to the applications and guidance

Application 1. g is defined on $[1/2;+\infty[$ and differentiable on $]1/2;+\infty[$, with $g'(x) = 1/\sqrt{2x-1} > 0$. h is differentiable on $\mathbb{R}$, $h'(x) = -2xe^{1-x^2}$.

Application 2. The inner derivative is 3, so $f' = 12(3x+1)^3$. The coefficient x in the expansion of $(1+3x)^4$ is 12, hence $f'(0) = 12$, incompatible with the proposed formula.

Application 3. C is f, A is f' and B is f''. The zeros of A at ± 1 correspond to the extrema of C; the zero of B at 0 separates where A decreases and where it increases. Calculation: $f' = 3x^2 - 3$, $f'' = 6x$.

Application 4. All three are continuous. Polynomials are differentiable; $|x|$ has one-sided slopes -1 and 1. For x^4, $f'' = 12x^2 \geq 0$, so no inflection at 0; for x^3, $f'' = 6x$ changes sign, so inflection at 0. $|x|$ is convex on $\mathbb{R}$ and has no change of convexity at 0.

Application 5. The step function never takes the value 0: it is not continuous at 0. x^2 is continuous but not strictly monotonic on $\mathbb{R}$; $x^2 = 1$ has the two solutions ± 1.

Application 6. Let $G(x) = e^{-x} - x$, continuous, with $G(0) = 1 > 0$ and $G(1) = e^{-1} - 1 < 0$. $G' = -e^{-x} - 1 < 0$ ensures uniqueness. $G(1/2) > 0$, then $G(3/4) < 0$: interval $[1/2;3/4]$. From $[0;1]$, seven steps give a width of 1/128; six are not enough.

Scaffolding and review

Hide the formulas when first working on the three curves. If the pupil confuses sign and variation, have the pupil point to a portion of f negative but increasing. After a week, ask a new justification of uniqueness without the instruction "use the IVT".

Teaching notes and solutions

Guidance for the teacher

The progression packs this chapter into three sessions: genuinely split S29 between differentiation and convexity, then revisit the proof about tangents in S33. The vocabulary continuous/differentiable/convex must be distinguished using examples. Continuity gives existence by the IVT; the strict monotonicity gives uniqueness. For bisection, require an initial interval where the sign changes and a continuous function. A change of convexity, and not merely the vanishing of f'', characterises an inflection point.

Solutions to the activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero at -1, 1, negative between these values. The sign of f'' is negative to the left of 0, zero at 0, positive to right.

2. The variations and limits are summarised by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. Convexity changes at 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent has equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. On $]-\infty, -1]$, it is strictly increasing from $-\infty$ to 2: level 1 is attained exactly once. On $[-1, 1]$, it strictly decreases from 2 to -2: a second solution. On $[1, +\infty[$, it strictly increases from -2 to $+\infty$: a third one. The points -1, 1 are not not solutions, so no solution is counted twice.

5. Continuity guarantees a preimage for intermediate values. It does not give uniqueness: the function studied is continuous and the value 1 has three preimages.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are differentiable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside of $[-1, 1]$, negative inside. Hence f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. The changes of convexity give the inflection points $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve is above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Let $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. Moreover $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the IVT gives a zero in $]1, 2[$, and being strictly increasing on $\mathbb{R}$ ensures that it is unique.

E5. After n steps, the length is $\frac{2}{2^n}$. The requested condition is equivalent to $2^n \geq 2000$. Since $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps suffice and ten do not.

Solutions to homework DM4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. At $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparative growth. By product, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0$: f strictly decreases from 1 to 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. Hence f is concave on $[0,1]$, convex on $[1, +\infty[$. The change of convexity at 1 gives the inflection point $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is below $\frac{1}{2}$. The corollary of the intermediate value theorem gives a unique solution α. In addition, $f(1) = \frac{2}{e} \simeq 0{,}7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0{,}4060 < \frac{1}{2}$, therefore $1 < \alpha < 2$.

4.
```
from math import exp

def racine_dm4():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+1)*exp(-m) > 0.5:
            a = m
        else :
            b = m
    return a, b
```

The decrease preserves $f(a) \geq \frac{1}{2} \geq f(b)$; α remains in $[a,b]$. After eleven steps, we get $[1{,}677734375, 1{,}6787109375]$, of length $\frac{1}{1024} < 10^{-3}$. The value $\alpha \simeq 1{,}67835$ is compatible with these bounds.

5. The tangent at 0 is $y = f(0) + f'(0)x = 1$. The variations give $f(x) \leq 1$, so the whole curve is below this tangent. This does not characterise concavity: it would have to be below all the tangents, whereas f becomes convex after 1.

Solutions to the textbook or past-paper work

For $f(x) = 5x^2e^x$, differentiating a product gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $]-\infty;-2[$ and $]0;+\infty[$, negative on $]-2;0[$, zero at -2 & 0. The function therefore increases up to $f(-2) = 20e^{-2}$, decreases down to $f(0) = 0$, then increases. Its limits are 0 at $-\infty$ (comparative growth) and $+\infty$ at $+\infty$. Differentiating again gives

$$f''(x) = 5(x^2+4x+2)e^x = 5\big((x+2)^2-2\big)e^x.$$

Let $a = -2-\sqrt{2}$ and $b = -2+\sqrt{2}$. The polynomial factor is positive outside of $[a;b]$, negative in $]a;b[$. Thus f is convex on $]-\infty;a]$ and $[b;+\infty[$, concave on $[a;b]$. The sign changes at both roots: the inflection points are $(a;5a^2e^a)$ and $(b;5b^2e^b)$.

The variation table can be summarised as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	$\nearrow$	$20e^{-2}$	$\searrow$	0	$\nearrow$	$+\infty$

For question 5, the window below shows both changes of convexity and both extrema. The marked points correspond to the computed inflection points; the plot is used to check the consistency of the study.

